

# DISCRETE MATHEMATICS

## Contest 4 – MCQ Practice Exam

105 Questions • Two-Phase Distribution • With Explanations

**Phase 1 (40 Qs – 38%)**

**Phase 2 (65 Qs – 62%)**

Logic, Proofs, Set Theory

Abstract Algebra & Advanced Counting

**Instructions:** Each question has exactly one correct answer. Read all options before answering. Answers and brief explanations appear below each question. **Phase 2 carries more weight – prioritise it.**

## PHASE 1: Logic, Proofs & Set Theory

Q1–Q40 • 40 Questions • 38% Weightage

### Section A: Propositional Logic, Predicate Logic & Boolean Algebra Q1–Q14

**Q1.** Which of the following is logically equivalent to  $p \rightarrow q$ ?

(A)  $p \wedge \neg q$

(B)  $\neg p \vee q$

(C)  $\neg q \rightarrow p$

(D)  $q \rightarrow p$

**Answer: (B)**

7

An implication  $p \rightarrow q$  is equivalent to  $\neg p \vee q$ ; this is the disjunctive form of implication.

**Q2.** The contrapositive of  $p \rightarrow q$  is:

(A)  $q \rightarrow p$

(B)  $\neg p \rightarrow \neg q$

(C)  $\neg q \rightarrow \neg p$

(D)  $p \wedge \neg q$

**Answer: (C)**

7

The contrapositive of  $p \rightarrow q$  is  $\neg q \rightarrow \neg p$ , and it is logically equivalent to the original.

**Q3.** Under what condition is the implication  $p \rightarrow q$  **false**?

(A)  $p$  is false and  $q$  is true

(B)  $p$  is true and  $q$  is false

(C) Both  $p$  and  $q$  are false

(D) Both  $p$  and  $q$  are true

**Answer: (B)**

7

An implication is false only when the premise is true and the conclusion is false; all other cases yield true.

**Q4.** Which of the following is De Morgan's law applied to  $\neg(p \vee q)$ ?

- (A)  $\neg p \vee \neg q$  (B)  $\neg p \wedge \neg q$   
 (C)  $p \wedge q$  (D)  $p \vee \neg q$

**Answer: (B)**

7

De Morgan's law:  $\neg(p \vee q) \equiv \neg p \wedge \neg q$ . The negation distributes and OR flips to AND.

**Q5.** The biconditional  $p \leftrightarrow q$  is true when:

- (A)  $p$  is true and  $q$  is false (B)  $p$  and  $q$  have opposite truth values  
 (C)  $p$  and  $q$  have the same truth value (D)  $p$  is false only

**Answer: (C)**

7

A biconditional  $p \leftrightarrow q$  is true exactly when both have the same truth value (both T or both F).

**Q6.** Which quantifier negation is correct?

- (A)  $\neg(\forall x P(x)) \equiv \forall x \neg P(x)$  (B)  $\neg(\exists x P(x)) \equiv \exists x \neg P(x)$   
 (C)  $\neg(\forall x P(x)) \equiv \exists x \neg P(x)$  (D)  $\neg(\forall x P(x)) \equiv \forall x P(x)$

**Answer: (C)**

7

By De Morgan for predicates:  $\neg(\forall x P(x)) \equiv \exists x \neg P(x)$  – the quantifier switches and the predicate is negated.

**Q7.** The statement  $\forall x \exists y (x < y)$  over integers means:

- (A) There is a largest integer (B) Every integer has an integer smaller than it  
 (C) For every integer, there exists a larger integer (D) There exists an integer larger than all others

**Answer: (C)**

7

$\forall x \exists y (x < y)$  says: for each  $x$ , we can find a  $y$  that exceeds it – the integers have no maximum.

**Q8.** In Boolean algebra, the expression  $x \wedge \bar{x}$  equals:

- (A)  $x$  (B) 1  
 (C) 0 (D)  $\bar{x}$

**Answer: (C)**

7

A variable AND-ed with its complement always equals 0 (contradiction law).

**Q9.** Which of the following is an example of a tautology?

- (A)  $p \wedge \neg p$  (B)  $p \vee q$   
 (C)  $p \vee \neg p$  (D)  $p \rightarrow \neg p$

**Answer: (C)**

7

$p \vee \neg p$  is always true regardless of  $p$ 's value – this is the law of excluded middle, a tautology.

**Q10.** The order of nested quantifiers  $\forall x \exists y P(x, y)$  vs  $\exists y \forall x P(x, y)$ :

- (A) They are always logically equivalent (B) The second implies the first but not vice versa  
 (C) The first implies the second but not vice versa (D) They are equivalent only when  $P$  is symmetric

**Answer: (B)**

7

$\exists y \forall x P(x, y)$  (one fixed  $y$  works for all  $x$ ) is stronger and implies  $\forall x \exists y P(x, y)$ , but not conversely.

**Q11.** In SOP (Sum of Products) form, the expression is a:

- (A) Product of sums of literals (B) Sum (OR) of product (AND) terms  
 (C) Single AND of all variables (D) Negation of all minterms

**Answer: (B)**

7

Sum of Products = OR of AND-terms (minterms); it is a standard canonical form for Boolean expressions.

**Q12.** Which Boolean law states that  $x \wedge (x \vee y) \equiv x$ ?

- (A) Idempotent law (B) Absorption law  
 (C) Distributive law (D) Complement law

**Answer: (B)**

7

The absorption law:  $x \wedge (x \vee y) \equiv x$  and  $x \vee (x \wedge y) \equiv x$  – the extra term is “absorbed.”

**Q13.** If  $p$  is “It rains” and  $q$  is “The ground is wet,” which represents “If it does not rain, the ground is not wet”?

- (A)  $p \rightarrow q$  (B)  $\neg q \rightarrow \neg p$   
 (C)  $\neg p \rightarrow \neg q$  (D)  $q \rightarrow p$

**Answer: (C)**

7

“If not  $p$  then not  $q$ ” is  $\neg p \rightarrow \neg q$ , which is the *inverse* of  $p \rightarrow q$  (NOT equivalent to it).

**Q14.** Which of the following best describes a predicate?

- (A) A proposition with a fixed truth value      (B) A function that returns a truth value depending on its argument
- (C) A tautology involving two variables      (D) An expression with no variables

**Answer: (B)**

7

A predicate  $P(x)$  is a propositional function – it becomes a proposition (T or F) once  $x$  is assigned a value.

## Section B: Rules of Inference, Proof Types & Loop Invariants

*Q15–Q28*

**Q15.** Modus Ponens states: given  $p$  and  $p \rightarrow q$ , we can conclude:

- (A)**  $\neg p$

**(C)**  $q$

**(B)**  $\neg q$

**(D)**  $p \rightarrow \neg q$

**Answer: (C)**

7

Modus Ponens:  $p, p \rightarrow q \vdash q$ . If the premise is true and the implication holds, the conclusion follows.

**Q16.** Modus Tollens states: given  $\neg q$  and  $p \rightarrow q$ , we can conclude:

- (A)  $q$

(C)  $\neg p$

(B)  $p$

(D)  $p \wedge q$

**Answer: (C)**

7

Modus Tollens:  $\neg q, p \rightarrow q \vdash \neg p$ . If the conclusion is false and the implication holds, the premise must be false.

**Q17.** In a proof by contradiction, we:

- (A) Prove the statement directly from axioms  
(B) Assume the negation and derive a contradiction  
(C) Prove the contrapositive  
(D) Split into exhaustive cases

**Answer: (B)**

7

Proof by contradiction (reductio ad absurdum): assume  $\neg p$ , derive  $\perp$  (falsehood), conclude  $p$  must be true.

**Q18.** To prove “ $\sqrt{2}$  is irrational” by contradiction, we first assume:

- (A)  $\sqrt{2} = p/q$  in lowest terms where  $p, q$  are integers  
 (B)  $\sqrt{2}$  is rational, i.e.  $\sqrt{2} = p/q$  with  $\gcd(p, q) = 1$   
 (C) 2 is not a perfect square  
 (D)  $p^2 = 2q^2$  has no solutions

**Answer: (B)**

7

We assume  $\sqrt{2} = p/q$  in lowest terms (both not even), then show both must be even – contradiction.

**Q19.** Hypothetical Syllogism allows us to conclude from  $p \rightarrow q$  and  $q \rightarrow r$ :

- (A)  $p \rightarrow r$   
 (B)  $p \wedge r$   
 (C)  $q \rightarrow p$   
 (D)  $r \rightarrow p$

**Answer: (A)**

7

Hypothetical Syllogism: if  $p \rightarrow q$  and  $q \rightarrow r$ , then  $p \rightarrow r$  (transitivity of implication).

**Q20.** Disjunctive Syllogism: from  $p \vee q$  and  $\neg p$ , we conclude:

- (A)  $p$   
 (B)  $\neg q$   
 (C)  $q$   
 (D)  $p \wedge q$

**Answer: (C)**

7

Disjunctive Syllogism: one disjunct is false, so the other must be true:  $p \vee q, \neg p \vdash q$ .

**Q21.** In proof by induction, the inductive step proves:

- (A) The base case holds  
 (B)  $P(k) \rightarrow P(k+1)$  for arbitrary  $k$   
 (C) All cases are exhaustive  
 (D)  $P(n)$  holds for  $n = 1$  only

**Answer: (B)**

7

The inductive step assumes  $P(k)$  (inductive hypothesis) and uses it to prove  $P(k+1)$ , completing the induction.

**Q22.** Which proof technique is most suitable to prove that  $n^2 + n$  is even for all positive integers  $n$ ?

- (A) Proof by contradiction  
 (B) Proof by contrapositive  
 (C) Direct proof or mathematical induction  
 (D) Proof by exhaustion of infinite cases

**Answer: (C)**

7

A direct proof (factor as  $n(n+1)$ , one factor is always even) or induction both work cleanly

here.

**Q23.** A proof by contrapositive of  $p \rightarrow q$  actually proves:

- (A)  $p \rightarrow \neg q$  (B)  $q \rightarrow p$   
 (C)  $\neg q \rightarrow \neg p$  (D)  $\neg p \rightarrow q$

**Answer: (C)**

7

The contrapositive  $\neg q \rightarrow \neg p$  is logically equivalent to  $p \rightarrow q$ , so proving it suffices.

**Q24.** Which of the following is a valid rule of inference?

- (A)  $p \rightarrow q, q \vdash p$  (Affirming the consequent) (B)  $p \rightarrow q, \neg p \vdash \neg q$  (Denying the antecedent)  
 (C)  $p \vee q, \neg p \vdash q$  (Disjunctive Syllogism) (D)  $p \wedge q \vdash p \rightarrow q$

**Answer: (C)**

7

Options (A) and (B) are logical fallacies. Disjunctive Syllogism (C) is a valid rule of inference.

**Q25.** A loop invariant must hold at which points?

- (A) Only before the loop begins (B) Before, during (each iteration), and after the loop  
 (C) Only after the loop terminates (D) Only during odd-numbered iterations

**Answer: (B)**

7

A loop invariant is established at initialization, maintained through each iteration, and used at termination to prove correctness.

**Q26.** In proving loop correctness, the **termination** step shows:

- (A) The invariant holds before the loop (B) The invariant is preserved each iteration  
 (C) The loop eventually ends and the invariant gives the desired result (D) The loop runs in  $O(n)$  time

**Answer: (C)**

7

Termination combines the loop invariant with the exit condition to prove the program achieves its specification.

**Q27.** Which statement is true about proof by cases?

- (A) It requires infinitely many cases to be valid  
 (B) The cases must be mutually exclusive and collectively exhaustive  
 (C) It is weaker than direct proof  
 (D) It cannot handle more than 3 cases

**Answer: (B)**

7

Proof by cases divides the domain into exhaustive (covering all possibilities) and often mutually exclusive cases.

**Q28.** To prove  $P(n)$  for all  $n \geq 3$  using induction, the base case should verify:

- (A)  $P(1)$   
 (B)  $P(2)$   
 (C)  $P(3)$   
 (D)  $P(0)$  and  $P(1)$

**Answer: (C)**

7

When the claim is “for all  $n \geq 3$ ,” the base case is  $P(3)$ , the smallest value in the domain.

### Section C: Set Theory, Countability & Cantor’s Diagonal

Q29–Q40

**Q29.** Russell’s Paradox arises when we consider the set:

- (A)  $S = \{x : x \in x\}$   
 (B)  $S = \{x : x \notin x\}$   
 (C)  $S = \{x : x = x\}$   
 (D)  $S = \{x : |x| < \infty\}$

**Answer: (B)**

7

$S = \{x : x \notin x\}$  leads to the contradiction: if  $S \in S$  then  $S \notin S$ , and vice versa.

**Q30.** What does Russell’s Paradox motivate in mathematics?

- (A) Abandoning set theory altogether  
 (B) The need for axiomatic set theory (like ZFC) with restricted comprehension  
 (C) Allowing sets to contain themselves freely  
 (D) Using only finite sets

**Answer: (B)**

7

Russell’s Paradox shows that naive comprehension (any property defines a set) is inconsistent; ZFC restricts set formation.

**Q31.** A set  $A$  is countably infinite if:

- (A) It has exactly  $n$  elements for some  $n$   
 (B) There exists a bijection  $f : \mathbb{N} \rightarrow A$   
 (C) It contains only rational numbers  
 (D) It is a subset of  $\mathbb{Z}$

**Answer: (B)**

7

Countably infinite means there is a one-to-one correspondence (bijection) between  $\mathbb{N}$  and  $A$ .

**Q32.** Which of the following sets is **uncountable**?

- (A)  $\mathbb{Z}$  (integers) (B)  $\mathbb{Q}$  (rationals)  
 (C)  $\mathbb{N} \times \mathbb{N}$  (D)  $\mathbb{R}$  (reals)

**Answer: (D)**

7

$\mathbb{R}$  is uncountable (proved by Cantor's diagonal argument).  $\mathbb{Z}$ ,  $\mathbb{Q}$ , and  $\mathbb{N} \times \mathbb{N}$  are all countable.

**Q33.** Cantor's diagonal argument proves that  $[0, 1]$  is uncountable by:

- (A) Showing it has more elements than  $\mathbb{N}$  (B) Constructing a real number that differs from every listed real in at least one digit  
 (C) Mapping  $[0, 1]$  onto  $\mathbb{R}$  (D) Showing  $[0, 1]$  contains irrational numbers

**Answer: (B)**

7

The diagonal argument builds a number  $d$  where  $d_i \neq$  the  $i$ -th digit of  $r_i$ , so  $d$  differs from every listed number.

**Q34.** The power set  $\mathcal{P}(\mathbb{N})$  is:

- (A) Countably infinite (B) Finite  
 (C) Uncountable (D) Equal in size to  $\mathbb{N}$

**Answer: (C)**

7

By Cantor's theorem,  $|\mathcal{P}(A)| > |A|$  for any set  $A$ ; thus  $\mathcal{P}(\mathbb{N})$  is strictly larger than  $\mathbb{N}$  and hence uncountable.

**Q35.** The set of all programs (strings over a finite alphabet) is:

- (A) Uncountable (B) Countably infinite  
 (C) Finite (D) In bijection with  $\mathbb{R}$

**Answer: (B)**

7

Programs are finite strings over a finite alphabet; the set of all finite strings over a finite alphabet is countable.

**Q36.** Why does the existence of uncomputable functions follow from countability arguments?

- (A) All functions are computable                      (B) Programs are countable but functions  $\mathbb{N} \rightarrow \mathbb{N}$  are uncountable, so most functions have no program
- (C) Functions are countable and programs are uncountable                      (D) The halting problem has a finite solution

**Answer: (B)**

7

Since programs are countable and functions  $\mathbb{N} \rightarrow \mathbb{N}$  are uncountable, most functions cannot be matched to any program.

**Q37.** Which of the following is a bijection demonstrating that  $\mathbb{Z}$  is countable?

- (A)  $f(n) = n^2$                       (B)  $f(n) = |n|$
- (C) The mapping  $0, 1, -1, 2, -2, 3, -3, \dots$  lists all integers                      (D)  $f(n) = n/2$  for even  $n$

**Answer: (C)**

7

Listing  $0, 1, -1, 2, -2, \dots$  provides a bijection from  $\mathbb{N}$  to  $\mathbb{Z}$ , proving  $\mathbb{Z}$  is countable.

**Q38.** The Halting Problem is undecidable because:

- (A) No one has been clever enough to solve it yet                      (B) It reduces to the countability of programs
- (C) Assuming a halting decider leads to a contradiction via diagonal argument                      (D) Turing machines are too slow

**Answer: (C)**

7

Assuming a halting decider  $H$  exists leads to a self-referential program that contradicts  $H$ 's output – a diagonal argument.

**Q39.** If  $A$  and  $B$  are countable sets, then  $A \cup B$  is:

- (A) Always uncountable                      (B) Countable
- (C) Uncountable if  $A \cap B = \emptyset$                       (D) Uncountable if both are infinite

**Answer: (B)**

7

A finite union of countable sets is countable; we can interleave the enumerations of  $A$  and  $B$ .

**Q40.** Which statement about  $\mathbb{Q}$  (the rationals) is true?

- (A)  $\mathbb{Q}$  is uncountable because it is dense in  $\mathbb{R}$       (B)  $\mathbb{Q}$  is countable because it can be listed in a grid  $\mathbb{Z} \times \mathbb{Z}^+$  and diagonalised  
(C)  $\mathbb{Q}$  has the same cardinality as  $\mathbb{R}$       (D)  $\mathbb{Q}$  is finite

**Answer: (B)**

7

The rationals can be arranged in an infinite grid  $p/q$  and enumerated by Cantor's diagonal traversal, proving countability.

## PHASE 2: Abstract Algebra & Advanced Counting

Q41–Q105 • 65 Questions • 62% Weightage

### Section D: Group Theory

Q41–Q65

**Q41.** Which of the following is NOT a group axiom?

- (A) Closure      (B) Commutativity  
(C) Existence of identity      (D) Existence of inverses

**Answer: (B)**

7

Commutativity is the property of *abelian* groups – it is not required for a general group (CAII: Closure, Associativity, Identity, Inverse).

**Q42.** The group  $(\mathbb{Z}_6, +_6)$  has order:

- (A) 3      (B) 6  
(C) 12      (D) Infinite

**Answer: (B)**

7

$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$  has 6 elements, so its order is 6.

**Q43.** In  $(\mathbb{Z}_8, +_8)$ , what is the order of element 2?

- (A) 2      (B) 4  
(C) 8      (D) 1

**Answer: (B)**

7

$2 + 2 = 4$ ,  $4 + 2 = 6$ ,  $6 + 2 = 0$ . It takes 4 steps to reach the identity 0, so  $\text{ord}(2) = 4$ .

**Q44.** Which of the following is an abelian group?

- (A)  $S_3$  (symmetric group on 3 elements)    (B)  $(\mathbb{Z}, +)$   
 (C)  $(\text{GL}_2(\mathbb{R}), \times)$  (invertible  $2 \times 2$  matrices)    (D)  $S_n$  for  $n \geq 3$

**Answer: (B)**

7

$(\mathbb{Z}, +)$  is abelian since  $a + b = b + a$  for all integers.  $S_n$  is non-abelian for  $n \geq 3$ .

**Q45.** The inverse of element 5 in  $(\mathbb{Z}_7, +_7)$  is:

- (A) 5    (B) 2  
 (C) 1    (D) 6

**Answer: (B)**

7

We need  $5 + x \equiv 0 \pmod{7}$ , so  $x = 7 - 5 = 2$ . Thus  $5^{-1} = 2$  in  $\mathbb{Z}_7$ .

**Q46.** By Lagrange's Theorem, if  $|G| = 12$  and  $H \leq G$ , which cannot be  $|H|$ ?

- (A) 4    (B) 5  
 (C) 6    (D) 3

**Answer: (B)**

7

Lagrange's Theorem:  $|H|$  must divide  $|G| = 12$ . Since  $5 \nmid 12$ , a subgroup of order 5 cannot exist.

**Q47.** What does Lagrange's Theorem state?

- (A) Every group has a subgroup of every order    (B) The order of a subgroup divides the order of the group  
 (C) All subgroups are normal    (D) The order of any element equals the group order

**Answer: (B)**

7

Lagrange's Theorem: if  $H$  is a subgroup of finite group  $G$ , then  $|H|$  divides  $|G|$ .

**Q48.** If  $|G| = p$  for prime  $p$ , then  $G$  is:

- (A) Always non-abelian    (B) Cyclic and every non-identity element is a generator  
 (C) Has subgroups of every order from 1 to  $p$     (D) Not guaranteed to be abelian

**Answer: (B)**

7

By Lagrange, subgroup orders divide  $p$  (prime), so subgroups have order 1 or  $p$  only; every non-identity element generates all of  $G$ .

**Q49.** The number of generators of  $\mathbb{Z}_{12}$  is:

- (A) 4 (B) 6  
(C) 3 (D) 12

**Answer: (A)**

7

Generators of  $\mathbb{Z}_n$  are elements with  $\gcd(a, n) = 1$ .  $\varphi(12) = 12 \cdot (1 - 1/2) \cdot (1 - 1/3) = 4$ ; generators are  $\{1, 5, 7, 11\}$ .

**Q50.** A cyclic group  $\langle g \rangle$  of order  $n$  has  $\text{ord}(g^k) =$ :

- (A)  $k$  (B)  $n/k$   
(C)  $n/\gcd(n, k)$  (D)  $\gcd(n, k)$

**Answer: (C)**

7

The order of  $g^k$  in a cyclic group of order  $n$  is  $n/\gcd(n, k)$ .

**Q51.** Which of the following is cyclic?

- (A)  $S_3$  (symmetric group on 3 elements) (B)  $(\mathbb{Z}_{10}, +_{10})$   
(C)  $(\mathbb{Z}_2 \times \mathbb{Z}_2, +)$  (D)  $(\text{GL}_2(\mathbb{F}_2), \times)$

**Answer: (B)**

7

$\mathbb{Z}_{10}$  is cyclic, generated by 1 (or any element with  $\gcd(a, 10) = 1$ ). Every cyclic group is abelian.

**Q52.** The cosets of  $H = \{0, 4\}$  in  $\mathbb{Z}_8$  are:

- (A)  $\{0, 4\}, \{1, 5\}, \{2, 6\}, \{3, 7\}$  (B)  $\{0, 4\}, \{2, 6\}, \{4, 8\}$   
(C)  $\{0, 2, 4, 6\}, \{1, 3, 5, 7\}$  (D)  $\{0, 4, 8\}$  only

**Answer: (A)**

7

$|H| = 2$ ,  $|\mathbb{Z}_8| = 8$ , so there are  $8/2 = 4$  cosets:  $0 + H = \{0, 4\}$ ,  $1 + H = \{1, 5\}$ ,  $2 + H = \{2, 6\}$ ,  $3 + H = \{3, 7\}$ .

**Q53.** In  $\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}$  under multiplication mod 7, what is the order of element 3?

- (A) 3 (B) 6  
(C) 4 (D) 2

**Answer: (B)**

7

$3^1 = 3, 3^2 = 2, 3^3 = 6, 3^4 = 4, 3^5 = 5, 3^6 = 1 \pmod{7}$ . Order is 6, so 3 is a primitive root mod 7.

**Q54.** Euler's Theorem states that if  $\gcd(a, n) = 1$ , then:

- (A)  $a^n \equiv 1 \pmod{n}$  (B)  $a^{n-1} \equiv 1 \pmod{n}$   
 (C)  $a^{\varphi(n)} \equiv 1 \pmod{n}$  (D)  $a^2 \equiv 1 \pmod{n}$

**Answer: (C)**

7

Euler's Theorem:  $a^{\varphi(n)} \equiv 1 \pmod{n}$  when  $\gcd(a, n) = 1$ . Fermat's little theorem is the special case  $n = p$  prime.

**Q55.**  $\varphi(p^k)$  for prime  $p$  equals:

- (A)  $p^k - 1$  (B)  $p^{k-1}(p - 1)$   
 (C)  $p^k(1 - 1/p)$  only for  $k = 1$  (D)  $(p - 1)^k$

**Answer: (B)**

7

$\varphi(p^k) = p^{k-1}(p - 1)$  by the formula for prime powers; this equals  $p^k - p^{k-1}$ .

**Q56.** Fermat's Little Theorem says that for prime  $p$  and  $\gcd(a, p) = 1$ :

- (A)  $a^p \equiv a \pmod{p}$  (B)  $a^{p+1} \equiv 1 \pmod{p}$   
 (C)  $a^{p-1} \equiv 1 \pmod{p}$  (D)  $a^p \equiv 0 \pmod{p}$

**Answer: (C)**

7

Fermat:  $a^{p-1} \equiv 1 \pmod{p}$ . Equivalently,  $a^p \equiv a \pmod{p}$  holds for all  $a$  (including when  $p|a$ ).

**Q57.** Which of the following is the correct Euler totient  $\varphi(12)$ ?

- (A) 6 (B) 4  
 (C) 8 (D) 3

**Answer: (B)**

7

$\varphi(12) = 12(1 - \frac{1}{2})(1 - \frac{1}{3}) = 12 \cdot \frac{1}{2} \cdot \frac{2}{3} = 4$ .

**Q58.** The number of distinct left cosets of  $H$  in  $G$  is called:

- (A) The order of  $H$  (B) The index  $[G : H]$   
 (C) The kernel of  $G$  (D) The center of  $G$

**Answer: (B)**

7

The index  $[G : H]$  is the number of distinct left cosets; Lagrange's theorem gives  $|G| =$

$$|H| \cdot [G : H].$$

**Q59.** A group  $G$  with  $|G| = 1$  is called:

- (A) Cyclic (B) The trivial group  
(C) Non-abelian (D) An integral domain

**Answer: (B)**

7

The trivial group contains only the identity element and is the unique group of order 1.

**Q60.** In  $\mathbb{Z}_5^* = \{1, 2, 3, 4\}$ , which elements are primitive roots?

- (A)  $\{1, 4\}$  (B)  $\{2, 3\}$   
(C)  $\{2, 4\}$  (D)  $\{1, 2, 3, 4\}$

**Answer: (B)**

7

$|\mathbb{Z}_5^*| = 4$ . Number of primitive roots  $= \varphi(4) = 2$ .  $g = 2$ : powers are 2, 4, 3, 1.  $g = 3$ : powers are 3, 4, 2, 1. So  $\{2, 3\}$ .

**Q61.** Every subgroup of a cyclic group is:

- (A) Non-abelian (B) Cyclic  
(C) Of prime order (D) Isomorphic to  $\mathbb{Z}$

**Answer: (B)**

7

Theorem: every subgroup of a cyclic group is cyclic. This is a fundamental result in group theory.

**Q62.** What is the order of element 4 in  $(\mathbb{Z}_{12}, +_{12})$ ?

- (A) 4 (B) 6  
(C) 3 (D) 12

**Answer: (C)**

7

$4 + 4 = 8$ ,  $8 + 4 = 0 \pmod{12}$ . Three steps to reach identity;  $\text{ord}(4) = 3$ . Check:  $12 / \gcd(12, 4) = 12/4 = 3$ .

**Q63.** Which of the following statements about cosets is FALSE?

- (A) Two left cosets are either equal or disjoint (B) All cosets have the same cardinality as  $H$   
(C) Left cosets always equal right cosets (D) Cosets partition  $G$

**Answer: (C)**

7

Left and right cosets coincide only in abelian groups (or normal subgroups); in general  $aH \neq Ha$ .

**Q64.** The identity element in  $(\mathbb{Z}_7^*, \times_7)$  is:

- (A) 0 (B) 7  
(C) 1 (D) 6

**Answer: (C)**

7

Under multiplication, the identity is 1, since  $1 \times a = a$  for all  $a$ .

**Q65.** In a group  $G$ , if  $a^5 = e$  and  $a^3 = e$ , then:

- (A)  $a^{15} = e$  only (B)  $a = e$  (identity)  
(C)  $a^2 = e$  (D)  $a^8 = e$  only

**Answer: (B)**

7

$\text{ord}(a)$  divides both 5 and 3;  $\gcd(5, 3) = 1$ , so  $\text{ord}(a) = 1$ , meaning  $a = e$ .

### Section E: Rings, Polynomial Rings & Fields

Q66–Q78

**Q66.** Which of the following is NOT required for  $(R, +, \times)$  to be a ring?

- (A)  $(R, +)$  is an abelian group (B) Multiplication is commutative  
(C) Multiplication is associative (D) Distributive laws hold

**Answer: (B)**

7

A ring requires an abelian additive group, associative multiplication, and distributivity – commutativity of multiplication is extra.

**Q67.**  $\mathbb{Z}_n$  is an integral domain if and only if:

- (A)  $n$  is even (B)  $n$  is a perfect square  
(C)  $n$  is prime (D)  $n > 10$

**Answer: (C)**

7

$\mathbb{Z}_n$  is an integral domain (no zero divisors) if and only if  $n$  is prime; in that case  $\mathbb{Z}_n$  is actually a field.

**Q68.** A field is a commutative ring with unity where:

- (A) Only addition has inverses                      (B) Every nonzero element has a multiplicative inverse  
(C) Multiplication is not associative              (D) The additive group is not required

**Answer: (B)**

7

A field requires every nonzero element to have a multiplicative inverse;  $\mathbb{Q}$ ,  $\mathbb{R}$ ,  $\mathbb{C}$ , and  $\mathbb{Z}_p$  (prime  $p$ ) are fields.

**Q69.** The polynomial  $x^2 + x + 1$  over  $\mathbb{Z}_2$  is:

- (A) Reducible, since it has a root in  $\mathbb{Z}_2$               (B) Irreducible over  $\mathbb{Z}_2$   
(C) Equal to  $x^2 - x - 1$                                   (D) Reducible as  $(x + 1)^2$  over  $\mathbb{Z}_2$

**Answer: (B)**

7

Check:  $f(0) = 1 \neq 0$ ,  $f(1) = 3 \equiv 1 \neq 0$  in  $\mathbb{Z}_2$ . No roots, so irreducible (degree 2 polynomial with no roots is irreducible).

**Q70.** In  $F[x]$  for a field  $F$ , the division algorithm guarantees:

- (A) Every polynomial is irreducible                      (B) For  $f, g \neq 0$ , we can write  $f = qg + r$  with  $\deg(r) < \deg(g)$   
(C) Polynomials cannot be factored                      (D) The degree of a product is unpredictable

**Answer: (B)**

7

$F[x]$  is a Euclidean domain; polynomial long division gives  $f = qg + r$  with remainder of strictly smaller degree.

**Q71.**  $\text{GF}(4)$  (the field with 4 elements) is constructed as:

- (A)  $\mathbb{Z}_4$                                                               (B)  $\mathbb{Z}_2[x]/\langle x^2 + x + 1 \rangle$   
(C)  $\mathbb{Z}_2 \times \mathbb{Z}_2$                                                       (D)  $\mathbb{Z}_2[x]/\langle x^2 \rangle$

**Answer: (B)**

7

$\text{GF}(4) = \mathbb{Z}_2[x]/\langle x^2 + x + 1 \rangle$  since  $x^2 + x + 1$  is irreducible over  $\mathbb{Z}_2$ ; this gives the unique field of order  $2^2 = 4$ .

**Q72.** A finite field must have order:

- (A) Any positive integer  $n$                                   (B)  $p^n$  for prime  $p$  and  $n \geq 1$   
(C) A prime  $p$  only                                              (D) An even number

**Answer: (B)**

7

By the classification theorem of finite fields, the order must be a prime power  $p^n$ ; there is a unique field of each such order.

**Q73.**  $\mathbb{Z}[x]$  (polynomials over integers) is:

- |                |                                                   |
|----------------|---------------------------------------------------|
| (A) A field    | (B) A commutative ring with unity but not a field |
| (C) Not a ring | (D) A group under multiplication                  |

**Answer: (B)**

7

$\mathbb{Z}[x]$  is a commutative ring with unity (the constant polynomial 1), but e.g.  $x$  has no multiplicative inverse, so it is not a field.

**Q74.** The hierarchy from least to most structure is:

- |                                          |                                                                    |
|------------------------------------------|--------------------------------------------------------------------|
| (A) Field $\subset$ Ring $\subset$ Group | (B) Group $\subset$ Ring $\subset$ Integral Domain $\subset$ Field |
| (C) Ring $\subset$ Group $\subset$ Field | (D) Integral Domain $\subset$ Ring $\subset$ Field $\subset$ Group |

**Answer: (B)**

7

Every field is an integral domain, every integral domain is a ring, every ring has an underlying additive group: Group  $\subset$  Ring  $\subset$  ID  $\subset$  Field.

**Q75.** The multiplicative group  $\mathbb{F}_q^*$  of a finite field is:

- |                                 |                                     |
|---------------------------------|-------------------------------------|
| (A) Always non-abelian          | (B) A cyclic group of order $q - 1$ |
| (C) A cyclic group of order $q$ | (D) Not always a group              |

**Answer: (B)**

7

The multiplicative group  $\mathbb{F}_q^*$  is cyclic of order  $q - 1$ ; this is why primitive elements (generators) always exist.

**Q76.** The number of primitive elements (primitive roots) in  $\mathbb{F}_q^*$  is:

- |                      |                  |
|----------------------|------------------|
| (A) $q$              | (B) $q - 1$      |
| (C) $\varphi(q - 1)$ | (D) $\varphi(q)$ |

**Answer: (C)**

7

Since  $\mathbb{F}_q^*$  is cyclic of order  $q - 1$ , the number of generators equals  $\varphi(q - 1)$  by the formula for cyclic groups.

**Q77.** An element  $g$  is a primitive root mod  $p$  if:

- (A)  $g^2 \equiv 1 \pmod{p}$  (B)  $g^k$  for  $k = 1, \dots, p-1$  generates all of  $\{1, \dots, p-1\}$   
 (C)  $g \equiv 1 \pmod{p}$  (D)  $\gcd(g, p) = p$

**Answer: (B)**

7

A primitive root mod  $p$  has order  $p-1$ , so its powers cycle through all  $p-1$  nonzero residues modulo  $p$ .

**Q78.**  $x^2 + 1$  is irreducible over  $\mathbb{R}$  because:

- (A) It has degree 2 (B) It has no real roots ( $\sqrt{-1} \notin \mathbb{R}$ )  
 (C) It cannot be divided by  $x$  (D) Its leading coefficient is 1

**Answer: (B)**

7

$x^2 + 1 = 0$  has no real solutions; a degree-2 polynomial over a field is irreducible iff it has no roots in that field.

### Section F: Discrete Logarithm & Diffie-Hellman

Q79–Q87

**Q79.** In Diffie-Hellman key exchange with public prime  $p = 23$  and  $g = 5$ : if Alice's secret is  $a = 6$ , her public value  $A$  is:

- (A)  $5^6 \bmod 23 = 8$  (B)  $6^5 \bmod 23$   
 (C)  $23^5 \bmod 6$  (D)  $5 \times 6 \bmod 23$

**Answer: (A)**

7

Alice computes  $A = g^a \bmod p = 5^6 \bmod 23 = 15625 \bmod 23 = 8$ .

**Q80.** The Discrete Logarithm Problem (DLP) asks:

- (A) Compute  $g^x \bmod p$  given  $g, x, p$  (B) Find  $x$  given  $g^x \equiv h \pmod{p}$ , with  $g, h, p$  known  
 (C) Factor  $p$  into primes (D) Compute  $\gcd(g, p)$

**Answer: (B)**

7

The DLP is: given  $g, h, p$ , find  $x$  such that  $g^x \equiv h \pmod{p}$ . This is believed to be computationally hard for large  $p$ .

**Q81.** The security of Diffie-Hellman relies on:

- (A) The difficulty of prime factorisation      (B) The difficulty of computing discrete logarithms  
(C) The secrecy of the prime  $p$       (D) The difficulty of polynomial evaluation

**Answer: (B)**

7

DH security rests on the hardness of the Discrete Logarithm Problem; an adversary sees  $g^a$  and  $g^b$  but cannot easily find  $g^{ab}$ .

**Q82.** In DH, Alice and Bob both compute the shared key as:

- (A)  $g \cdot a \cdot b \bmod p$       (B)  $g^{ab} \bmod p$   
(C)  $a^b \bmod p$       (D)  $g^{a+b} \bmod p$

**Answer: (B)**

7

Alice computes  $B^a = g^{ab}$  and Bob computes  $A^b = g^{ab}$ ; both arrive at  $K = g^{ab} \bmod p$ .

**Q83.** Fast modular exponentiation (repeated squaring) computes  $a^n \bmod m$  in:

- (A)  $O(n)$  multiplications      (B)  $O(\sqrt{n})$  multiplications  
(C)  $O(\log n)$  multiplications      (D)  $O(n^2)$  multiplications

**Answer: (C)**

7

By writing  $n$  in binary and using repeated squaring,  $a^n \bmod m$  requires only  $O(\log n)$  multiplications.

**Q84.** What does an eavesdropper observe in Diffie-Hellman?

- (A)  $g, p, g^a \bmod p, g^b \bmod p$       (B) The shared key  $g^{ab} \bmod p$   
(C) The private keys  $a$  and  $b$       (D) Only  $p$

**Answer: (A)**

7

The eavesdropper sees the public parameters  $g, p$  and the transmitted values  $A = g^a$  and  $B = g^b$ , but NOT  $a, b$ , or  $K$ .

**Q85.** To compute  $5^6 \bmod 23$  by repeated squaring, we write  $6 = 110_2$  and compute:

- (A)  $5^1 \cdot 5^2 \cdot 5^3$       (B)  $5^4 \cdot 5^2$  (since  $6 = 4 + 2$ )  
(C)  $5^8 \cdot 5^{-2}$       (D)  $5^6 = 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5$  directly

**Answer: (B)**

7

$6 = 4 + 2$ , so  $5^6 = 5^4 \cdot 5^2$ . Compute:  $5^2 = 2$ ,  $5^4 = 4$ ,  $5^6 = 4 \cdot 2 = 8 \pmod{23}$ .

**Q86.** Which property makes  $g$  a valid generator in DH?

- (A)  $\gcd(g, p) > 1$  (B)  $g$  is a primitive root mod  $p$   
 (C)  $g < p/2$  (D)  $g$  is even

**Answer: (B)**

7

For DH to work correctly,  $g$  must be a primitive root mod  $p$  so that  $g^k$  for  $k = 1, \dots, p-1$  covers all nonzero residues.

**Q87.** The multiplicative inverse of 3 modulo 7 is:

- (A) 4 (B) 5  
 (C) 2 (D) 6

**Answer: (B)**

7

$3 \times 5 = 15 \equiv 1 \pmod{7}$ , so  $3^{-1} \equiv 5 \pmod{7}$ .

### Section G: Recurrence Relations

Q88–Q96

**Q88.** The characteristic equation of  $a_n = 5a_{n-1} - 6a_{n-2}$  is:

- (A)  $r^2 - 5r + 6 = 0$  (B)  $r^2 + 5r - 6 = 0$   
 (C)  $r - 5 + 6 = 0$  (D)  $5r^2 - 6r + 1 = 0$

**Answer: (A)**

7

The characteristic equation of  $a_n = c_1a_{n-1} + c_2a_{n-2}$  is  $r^2 - c_1r - c_2 = 0$ ; here  $r^2 - 5r + 6 = 0$ .

**Q89.** The Fibonacci recurrence  $F_n = F_{n-1} + F_{n-2}$  has characteristic roots:

- (A) 1 and 2 (B)  $\phi = \frac{1+\sqrt{5}}{2}$  and  $\hat{\phi} = \frac{1-\sqrt{5}}{2}$   
 (C) 2 and  $-1$  (D) 1 and  $-1$

**Answer: (B)**

7

The characteristic equation  $r^2 - r - 1 = 0$  gives  $r = (1 \pm \sqrt{5})/2$ : the golden ratio  $\phi$  and its conjugate  $\hat{\phi}$ .

**Q90.** If the characteristic equation has a repeated root  $r$ , the general solution is:

- (A)  $a_n = Ar^n$  (B)  $a_n = (A + Bn)r^n$   
 (C)  $a_n = Ar^n + Br^{-n}$  (D)  $a_n = A \cos(rn) + B \sin(rn)$

**Answer: (B)**

7

For a repeated root  $r$  (multiplicity 2), the general solution is  $a_n = (A + Bn)r^n$ .

**Q91.** For the recurrence  $a_n = 3a_{n-1}$  with  $a_0 = 2$ , the closed form is:

- (A)  $a_n = 2n$  (B)  $a_n = 3^n$   
 (C)  $a_n = 2 \cdot 3^n$  (D)  $a_n = 2n \cdot 3^n$

**Answer: (C)**

7

The characteristic root is  $r = 3$ , so  $a_n = A \cdot 3^n$ . Using  $a_0 = 2$ :  $A = 2$ , giving  $a_n = 2 \cdot 3^n$ .

**Q92.** To solve a non-homogeneous recurrence  $a_n = 2a_{n-1} + 3^n$ , the particular solution trial is:

- (A)  $a_n^{(p)} = C \cdot 2^n$  (B)  $a_n^{(p)} = Cn$   
 (C)  $a_n^{(p)} = C \cdot 3^n$  (D)  $a_n^{(p)} = C \cdot n \cdot 3^n$

**Answer: (C)**

7

Since  $f(n) = 3^n$  and 3 is not a characteristic root of the homogeneous part (root is 2), try  $a_n^{(p)} = C \cdot 3^n$ .

**Q93.** Binary search satisfies  $T(n) = T(n/2) + 1$ . Its time complexity is:

- (A)  $O(n)$  (B)  $O(\log n)$   
 (C)  $O(\sqrt{n})$  (D)  $O(n \log n)$

**Answer: (B)**

7

By Master Theorem:  $a = 1, b = 2, n^{\log_2 1} = 1 = \Theta(1) = f(n)$ . Case 2:  $T(n) = \Theta(\log n)$ .

**Q94.** Merge Sort satisfies  $T(n) = 2T(n/2) + n$ . Its time complexity is:

- (A)  $O(n)$  (B)  $O(n^2)$   
 (C)  $O(n \log n)$  (D)  $O(\log n)$

**Answer: (C)**

7

Master Theorem:  $a = 2, b = 2, n^{\log_2 2} = n = \Theta(n) = f(n)$ . Case 2 ( $k=0$ ):  $T(n) = \Theta(n \log n)$ .

**Q95.** The closed form of the Fibonacci sequence is:

- (A)  $F_n = n/\sqrt{5}$  (B)  $F_n = \frac{\phi^n - \hat{\phi}^n}{\sqrt{5}}$  where  $\phi = \frac{1+\sqrt{5}}{2}$   
 (C)  $F_n = 2^{n-1}$  (D)  $F_n = n^2/5$

**Answer: (B)**

7

Binet's formula:  $F_n = (\phi^n - \hat{\phi}^n)/\sqrt{5}$ . This is derived by solving the characteristic equation and applying initial conditions.

**Q96.** For non-homogeneous  $a_n = a_{n-1} + d$  (constant  $d$ ), the particular solution trial is:

- (A)  $a_n^{(p)} = Cn^2$  (B)  $a_n^{(p)} = C$  (constant)  
 (C)  $a_n^{(p)} = Cn$  (linear) (D)  $a_n^{(p)} = Cd^n$

**Answer: (C)**

7

Since the characteristic root is 1 (from  $a_n - a_{n-1} = 0$ ) and  $f(n) = d = d \cdot 1^n$ , 1 is a root, so multiply by  $n$ : try  $Cn$ .

### Section H: Master Theorem & Complexity

Q97–Q101

**Q97.** For  $T(n) = 7T(n/2) + n^2$  (Strassen's algorithm), Master Theorem gives:

- (A)  $\Theta(n^2)$  (B)  $\Theta(n^2 \log n)$   
 (C)  $\Theta(n^{\log_2 7})$  (D)  $\Theta(n^3)$

**Answer: (C)**

7

$n^{\log_2 7} \approx n^{2.807}$ . Since  $f(n) = n^2 = O(n^{2.807-\epsilon})$ , Case 1 applies:  $T(n) = \Theta(n^{\log_2 7})$ .

**Q98.** Which is the correct ordering of complexity classes (slowest to fastest growth)?

- (A)  $O(n!) < O(2^n) < O(n^2) < O(1) < O(\log n) < O(\sqrt{n}) < O(n) < O(n \log n) < O(n) < O(\log n)$   
 (B)  $O(1) < O(\log n) < O(\sqrt{n}) < O(n) < O(n \log n) < O(n^2) < O(2^n) < O(n!)$   
 (C)  $O(\log n) < O(1) < O(n) < O(2^n)$  (D)  $O(n^2) < O(n \log n) < O(n) < O(\log n)$

**Answer: (B)**

7

The correct hierarchy from slowest to fastest:  $O(1), O(\log n), O(\sqrt{n}), O(n), O(n \log n), O(n^2), O(2^n), O(n!)$

**Q99.** For  $T(n) = 4T(n/2) + n^2$ , which Master Theorem case applies?

- (A) Case 1:  $T(n) = \Theta(n^2)$  (B) Case 2:  $T(n) = \Theta(n^2 \log n)$   
 (C) Case 3:  $T(n) = \Theta(n^3)$  (D) Case 1:  $T(n) = \Theta(n^3)$

**Answer: (B)**

7

$n^{\log_2 4} = n^2$ .  $f(n) = n^2 = \Theta(n^2 \log^0 n)$ . Case 2 with  $k = 0$ :  $T(n) = \Theta(n^2 \log n)$ .

**Q100.**  $f = \Theta(g)$  means:

- (A)  $f$  grows strictly faster than  $g$  (B)  $f$  grows no faster than  $g$   
 (C)  $f$  and  $g$  have the same asymptotic growth rate (D)  $g$  grows strictly faster than  $f$

**Answer: (C)**

7

$\Theta$  notation (tight bound):  $f = \Theta(g)$  means there exist constants  $c_1, c_2, N$  such that  $c_1g(n) \leq f(n) \leq c_2g(n)$  for all  $n > N$ .

**Q101.** For  $T(n) = 3T(n/9) + \sqrt{n}$ , applying the Master Theorem:

- (A) Case 1:  $T(n) = \Theta(n^{1/3})$  (B) Case 2:  $T(n) = \Theta(\sqrt{n} \log n)$   
 (C) Case 3:  $T(n) = \Theta(\sqrt{n})$  (D) Cannot apply the theorem

**Answer: (B)**

7

$n^{\log_9 3} = n^{1/2} = \sqrt{n}$ .  $f(n) = \sqrt{n} = \Theta(n^{1/2} \log^0 n)$ . Case 2 ( $k=0$ ):  $T(n) = \Theta(\sqrt{n} \log n)$ .

### Section I: Generating Functions

Q102–Q105

**Q102.** The ordinary generating function of the sequence  $a_n = 1$  for all  $n \geq 0$  is:

- (A)  $e^x$  (B)  $\frac{1}{1-x}$   
 (C)  $\frac{1}{1+x}$  (D)  $\ln(1-x)$

**Answer: (B)**

7

$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$  for  $|x| < 1$ . This is the geometric series – the most fundamental generating function.

**Q103.** The generating function for  $a_n = 2^n$  is:

- (A)  $\frac{1}{1-2x}$  (B)  $\frac{2}{1-x}$   
 (C)  $\frac{1}{1-x^2}$  (D)  $e^{2x}$

**Answer: (A)**

7

$\sum_{n=0}^{\infty} 2^n x^n = \sum_{n=0}^{\infty} (2x)^n = \frac{1}{1-2x}$  (geometric series with ratio  $2x$ ).

**Q104.** To solve the recurrence  $a_n = 3a_{n-1}$  using generating functions, we multiply by  $x^n$  and sum to get:

- (A)  $G(x) = \frac{a_0}{1-3x}$  (B)  $G(x) = \frac{3}{1-x}$   
 (C)  $G(x) = a_0(1-3x)$  (D)  $G(x) = a_0 \cdot x$

**Answer: (A)**

7

After multiplying by  $x^n$ , summing, and solving:  $(1-3x)G(x) = a_0$ , so  $G(x) = a_0/(1-3x) = a_0 \sum 3^n x^n$ , giving  $a_n = a_0 \cdot 3^n$ .

**Q105.** The coefficient of  $x^r$  in  $\frac{1}{(1-x)^3}$  counts:

- (A)  $r!$  (B)  $\binom{r+2}{2}$   
(C)  $2^r$  (D)  $3^r$

**Answer: (B)**

7

$[x^r]\frac{1}{(1-x)^k} = \binom{r+k-1}{k-1}$ . For  $k = 3$ :  $\binom{r+2}{2}$  – this counts the ways to choose  $r$  items from 3 types with repetition.

## QUICK REFERENCE SUMMARY

### Phase 1 Key Formulas – Logic, Proofs, Sets

**Implication:**  $p \rightarrow q \equiv \neg p \vee q$     **Contrapositive:**  $\neg q \rightarrow \neg p$  (equivalent)    **Converse:**  $q \rightarrow p$  (NOT equivalent)

**De Morgan:**  $\neg(p \wedge q) \equiv \neg p \vee \neg q$      $\neg(p \vee q) \equiv \neg p \wedge \neg q$

**Quantifier Negation:**  $\neg \forall x P(x) \equiv \exists x \neg P(x)$      $\neg \exists x P(x) \equiv \forall x \neg P(x)$

**Modus Ponens:**  $p, p \rightarrow q \vdash q$     **Modus Tollens:**  $\neg q, p \rightarrow q \vdash \neg p$

**Induction:** Base  $P(n_0)$  + Step  $P(k) \rightarrow P(k+1) \Rightarrow P(n)$  for all  $n \geq n_0$

**Russell's Paradox:**  $S = \{x : x \notin x\}$  leads to contradiction  $\Rightarrow$  motivates ZFC

**Countable:**  $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$     **Uncountable:**  $\mathbb{R}, [0, 1], \mathcal{P}(\mathbb{N})$

### Phase 2 Key Formulas – Abstract Algebra

**Group (CAII):** Closure + Associativity + Identity + Inverse. Abelian = + commutativity

**Lagrange:**  $|H| \mid |G|$  for  $H \leq G$ . Corollary:  $\text{ord}(a) \mid |G|$  and  $a^{|G|} = e$

**Cyclic generator of  $\mathbb{Z}_n$ :**  $\gcd(a, n) = 1$ . **Number of generators:**  $\varphi(n)$

$\text{ord}(g^k) = n / \gcd(n, k)$  in cyclic group of order  $n$

**Fermat:**  $a^{p-1} \equiv 1 \pmod{p}$  (prime  $p$ ,  $p \nmid a$ )

**Euler:**  $a^{\varphi(n)} \equiv 1 \pmod{n}$  ( $\gcd(a, n) = 1$ )     $\varphi(p) = p - 1$ ,  $\varphi(p^k) = p^{k-1}(p - 1)$

**Field  $\subset$  ID  $\subset$  Ring  $\subset$  Group** (structure hierarchy)

**DH:** Public  $(g, p)$ ; Alice sends  $g^a$ , Bob sends  $g^b$ ; shared key  $K = g^{ab} \pmod{p}$

**Primitive root mod  $p$ :**  $\varphi(p - 1)$  primitive roots exist;  $\mathbb{F}_q^*$  cyclic of order  $q - 1$

### Phase 2 Key Formulas – Advanced Counting

**Recurrence (homogeneous):** Char. eqn  $\rightarrow$  roots  $r_1, r_2$ :  $a_n = Ar_1^n + Br_2^n$

Repeated root  $r$ :  $a_n = (A + Bn)r^n$     Fibonacci:  $F_n = (\phi^n - \hat{\phi}^n) / \sqrt{5}$

**Master Theorem**  $T(n) = aT(n/b) + f(n)$ , compare  $f(n)$  vs  $n^{\log_b a}$ :

Case 1:  $f = O(n^{\log_b a - \epsilon})$      $T = \Theta(n^{\log_b a})$

Case 2:  $f = \Theta(n^{\log_b a} \log^k n)$      $T = \Theta(n^{\log_b a} \log^{k+1} n)$

Case 3:  $f = \Omega(n^{\log_b a + \epsilon})$  + regularity     $T = \Theta(f(n))$

**OGF shortcuts:**  $\frac{1}{1-x} = \sum x^n$      $\frac{1}{1-ax} = \sum a^n x^n$      $\frac{1}{(1-x)^k} = \sum \binom{n+k-1}{k-1} x^n$

**GF method:** Multiply recurrence by  $x^n$ , sum over  $n$ , solve for  $G(x)$ , partial fractions  $\rightarrow a_n$

Discrete Mathematics Contest 4 – 105 MCQs (Phase 1: Q1–40, Phase 2: Q41–105)